

TECH TO FEEL THE METAVERSE

TOUCH, TASTE, FEEL, HEAR AND MORE

Satvik Pandey | feedback@digit.in

The bulk of the conversation about the metaverse usually revolves around sight, and sometimes as an after-thought, sound.

We want to be able to experience a false reality visually, because we are by-and-large visual beings. However, what makes the real world feel so real is all of the supporting things we experience thanks to our other senses. A windy day is not just windy because we see things flapping about in the wind, or hear the sound of wind rustling. We do that bit decently well as of right now. What we cannot feel is the rush of wind on our skin, the force of the wind on our clothes, and if it's an especially strong wind, the laboured breathing and watery eyes. This is what ruins a good visual experience, even if the accompanying audio engineering is top notch.

And the audio engineering will need some serious work as well. Think about how you do a video meeting now. Everyone's voice comes through their microphone with the background noise of whatever is happening around them. And it's not spatial audio, because there's no need for it to be. Inside a metaverse, in, say, a large conference room, where you're seated around a table, each person has a position, and they have people close to them on their left and right, and people further away from them at the ends of the table or across from them. How useless is it to have the voice of your boss, who is seated on

your right, come into your headset at equal left and right balance? Just audio processing for each member of the team, to spatially place them and then give them all true spatial sound is a challenge to overcome, but thankfully a lot of work has already started. Some estimates suggest that the spatial audio market will be worth \$280 billion in just three years! And that's a place we're not starting from scratch. Can you even imagine how valuable a company that solves the problems of smell, feelings on skin and other things like inner ear balance and the like will be worth?

PAST EXPERIENCES

In 2003, Second Life allowed us to create an avatar with desired physical and psychological traits and live the life of our dreams. Quite literally. You could choose to teleport, fly, drive expensive cars (if you could design them), or just plant a field of carrots and walk amongst carrots all day... Your imagination and creativity was the only thing holding you back. The problem was the graphics weren't good, and the delays and slow response was terrible.

Games like GTA Online allowed you to do things such as heists, raid airbases, be chased by cops, and generally be the exact opposite of the real you. But it was scripted fun. Yes

there was a lot to do, and the attempt was to make it feel like total freedom, but it wasn't. Minecraft is a similar title, which might be a gateway to the metaverse, but with far less illegal things happening on screen. Targeted at kids, it is a fun world building experience that comes in handy for a future metaverse.

THE SENSES

Even though we have a long, long way to go before we can even be happy with the level of sight and sound we might experience in the metaverse, and servers are going to take a massive beating if they have to render and send out visual information about hundreds of players in a relatively small areas of the metaverse, those are still problems that are fixable, or can be worked around. In terms of sight and sound, we might be 40 or 50 per cent of the way towards mimicking real life. For the other senses however, we're essentially at 0 per cent, and that's where a lot of research is being focussed.

Smell

This might be the sense that we're most capable of replicating. Work was ongoing for a long time before the concept of metaverse was ever dreamed up. This is because people wanted to digitally create smell for a lot of marketing reasons. Imagine seeing an ad on TV about a bakery, and then smelling freshly baked goodies. Or imagine an ad for a beach holiday in the Maldives, and you get a whiff of sea salt and freshly cut coconuts... you'd be far more likely to buy the holiday package if you sensed it in more ways than visual, right?

Without a doubt.

And so people have been researching smell for some time now. In fact, a company called OVR technology is already selling scent devices to corporate and governments that attach on to VR headsets



Source: feelreal.com